Referee Review of Paper 01: Mean and Covariance Moderation Under Carryover

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Author. pmsimstats team

Purpose. This whitepaper is a referee review of paper 01, the comparison of the mean-moderation and covariance-moderation architectures under carryover. It reviews the three R Markdown sources in `analysis/report/01-dgp-mean-moderation-vs-mvn/`: the manuscript `report.Rmd` (2,292 lines), the standalone `appendices.Rmd`, and the supplementary key-points summary `bullets.Rmd`. It checks the reported numbers against the stored simulation output, the stated mechanisms against the code, and the manuscript against itself, and it ends with a recommended order of work.

Status note (2026-10-07). The review describes the manuscript as it stood on 2026-09-30. On 2026-10-07 the manuscript was restated (its goal is now the specification of the compendium's reference data-generating process): `report.Rmd` was rewritten (1,378 lines), Appendix B moved to `appendix-b.Rmd`, `appendices.Rmd` was removed, and the reviewed text was archived as `archive/report-2026-10-07-pre-restatement.Rmd`, to which the line numbers below refer approximately. A later referee report is in `revisions/referee-report-2026-10-07.md`. Section 2.1 still holds: the result files remain those of 2026-08-08, and the factorial has not been re-run under the current package (inspected, file dates and git log, 2026-10-07). The findings below have not been re-checked against the restated manuscript.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here. Symbols marked *quoted* are the manuscript's own, reproduced only to identify the passage under review; Section 5 recommends their replacement.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
B_i, b_i	biomarker; standardized biomarker	canonical
c_{bm}	covariance-moderation correlation of B_i with BR_{it}	canonical
β_{bm}	mean-moderation multiplier	canonical
BR_{it}, σ_{BR}	drug-response component; its SD	canonical
σ_{bm}	SD of the biomarker	canonical
D_{it}, $D_{bc,it}$	binary drug state; exposure-decayed drug indicator (code `Dbc`)	canonical
$\bar{D}_{bc}$	mean of $D_{bc,it}$ over occasions	local
$t_{1/2}$, λ, t_{sd}	carryover half-life; decay rate; time since discontinuation	canonical
$\beta_{bm:D}$, $\hat{\beta}_{bm:D}$	interaction estimand (`bm:Dbc`); its estimate	canonical
θ_{true}	calibrated true value, $-c_{bm}\sigma_{BR}$	canonical
κ	mean model-based SE (`betaSE`) over empirical SD of $\hat{\beta}_{bm:D}$	canonical
$\widehat{SE}$	model-based standard error of $\hat{\beta}_{bm:D}$	local
z	Wald or two-proportion z statistic	local
ρ	AR(1) serial correlation	canonical
g_t	coupling at occasion t: the fraction of c_{bm} applied (`docs/44`)	local
$\mu_{BR,1}$	BR mean at the first off-drug occasion of a run (manuscript Appendix B.6 example)	quoted
$\beta_1\beta_{bm}$	the manuscript's mean-moderation interaction product (Section 2.2.3)	quoted
ρ_c, c_1, $c_\times$	within-factor correlation; cross-factor correlation at the same and at different occasions (manuscript Appendix B.7)	quoted
b^{orig}	the manuscript's biomarker coupling vector (Appendix B)	quoted; collides with b_i
z_i, μ_B, σ_B, μ_{bm}, $\hat{\beta}_{bm:D_{bc}}$, p	non-canonical manuscript symbols discussed in Section 5	quoted

- **Mean moderation; covariance moderation.** The two data-generating architectures (NOTATION Part 3), formerly Architectures A and B.

- **Size-adjusted power.** Power with the critical value set at the 95th percentile of $|z|$ in the matching null cell.
- **Difference-of-differences z.** The z statistic for the difference between the two architectures' power losses.
- **CO, Hybrid, OL+BDC.** Crossover; the Hendrickson hybrid design; open-label titration followed by blinded discontinuation.

1. Evidence basis

Each finding below carries one of four evidence labels.

- **Verified:** the claim was recomputed or checked against data or code output in this review.
- **Inspected:** the relevant source code or document was read and the claim confirmed by reading, without execution.
- **Inferred:** the claim follows from verified facts but was not checked directly.
- **Unverified:** the claim is a hypothesis requiring a new run.

The following checks were performed.

- All relative losses, within-architecture z statistics, and difference-of-differences z statistics in Section 3.1 were recomputed by hand from the reported proportions. All match the manuscript. (Verified.)
- The Section 3.1 power tables were compared against `analysis/data/quick-sim/01-dgp-summary.txt`. All 18 alternative cells and all 18 null cells match. (Verified.)
- The replicate-level file `analysis/data/quick-sim/01-dgp-replicates.rds` (36,000 rows, with `beta`, `betaSE`, `p`) was reanalyzed for the ratio of mean model-based SE to empirical SD, and for size-adjusted power. (Verified; the script is reproduced in Appendix 1.)
- `R/generateData.R`, `R/lme_analysis.R`, and the driver `analysis/scripts/quick-sim/01-dgp-prototy` were read, and the git history of the package source and the result files was examined. (Inspected.)
- The rendered `report.pdf` table of contents was extracted with `pdftotext`. (Verified.)
- `tools/notation-lint.pl` was run over all three files. All three pass. The linter does not mechanize NOTATION rules 2 and 5, which are assessed manually in Section 5. (Verified.)

No new simulation was run. The size-adjusted power figures in Section 2.4 and the spacing-based design explanation in Section 2.5 are reanalyses of existing output and should be confirmed after the re-run recommended in Section 7.

2. Critical findings

These findings bear directly on the headline results, the stated mechanism, or both.

2.1 Section 3 results predate the carryover correction

Evidence: verified (file dates and git history).

The result files `01-dgp-summary.txt` and `01-dgp-replicates.rds` are dated 2026-08-08. The carryover recursion in `buildSigma()` was replaced by the anchored form in commit `955c604` of 2026-09-08 (`R/generateData.R:277-290`). Every number in Section 3 of the manuscript was therefore produced under the recursive form that Appendix B.6 describes as having been "corrected in September 2026".

Two passages are consequently inconsistent with each other and with the reported results.

- Section 2.2.4 (lines 577-579) states that the carryover line "in the current package is character-for-character the published one". That was true of the code that produced Section 3, but is no longer true of the package.
- Appendix B.6 and B.7 describe `mean` and `covar` as using the anchored form, and do not disclose that the reported power figures were not produced under it.

The corrected form retains more residual effect at the second and later off-drug occasions of each run (Appendix B.6's own worked example gives $0.25\,\mu_{BR,1}$ against $0.125\,\mu_{BR,1}$). The carryover-induced power losses are therefore likely to be larger under the current code, particularly in the designs with consecutive off-drug occasions (Hybrid and OL+BDC). (Inferred; the magnitude is unverified.)

Recommendation. Re-run the Section 3.1 factorial on the current package before any further revision of the Results, and state in the Reproducibility section the commit that produced the reported figures.

2.2 The conditional-variance-inflation mechanism is directionally inverted

Evidence: inspected (code) and verified (algebra).

Sections 2.2.3 and 3.2 attribute the architecture-specific loss to the identity

$$\mathrm{Var}(BR_{it} \mid B_i, D_{it}) = \sigma^2_{BR}(1 - c^2_{bm} D^2_{bc,it}),$$

arguing that "as carryover erodes $D_{bc,it}$ toward zero off drug" the conditional variance rises toward σ^2_{BR} and adds noise (lines 504-511, 841-847).

The direction of that argument is the reverse of what the implementation does.

- At $t_{1/2} = 0$, off-drug `Dbc` is set to exactly zero (`R/lme_analysis.R:112-113`), and the off-drug biomarker-response correlation is zero because `lambda_cor` is set to 0 and the decay branch is skipped (`R/generateData.R:225-228, 350`).
- For $t_{1/2} > 0$, off-drug `Dbc` equals $(1/2)^{t_{sd}/t_{1/2}} > 0$ after exposure (it is 0 at off-drug occasions before any exposure, `R/lme_analysis.R:114`).

Carryover therefore *raises* off-drug $D_{bc,it}$ from zero, and under the stated identity the off-drug conditional variance *falls* with carryover. Off-drug occasions are at the full unconditional variance σ^2_{BR} precisely in the no-carryover cell. The qualitative statement in Section 3.2 that the mechanism "needs sustained low-$D_{bc,it}$ time to accumulate" (line 907) has the same problem: low-$D_{bc,it}$ time is maximal at $t_{1/2} = 0$.

The identity itself is correct. What fails is the claim that carryover acts on power through it in the stated direction.

Recommendation. Withdraw the variance-inflation account as the mechanism, or re-derive it as a statement about the whole design (for example, about how the covariance-weighted information of the `bm:Dbc` contrast changes as off-drug occasions acquire intermediate exposure), and check the re-derivation against the replicate data before it is restored.

2.3 The Section 3.3 empirical check is misread

Evidence: verified (summary file and replicate reanalysis).

Section 3.3 states that covariance moderation's signal-to-noise erosion is "driven almost entirely by SE growth", and that "mean moderation shows no comparable SE trend" (lines 961-965). The signal-to-noise ratios quoted are arithmetically correct, being `mean_beta / sd_beta` from the summary file, but the interpretation does not hold.

Table 2: Empirical SD and mean $\hat{\beta}_{bm:D}$ by design and architecture as carryover increases

Design	Architecture	Empirical SD, $t_{1/2} = 0 \to 1$	Change	Mean $\hat{\beta}_{bm:D}$, $t_{1/2} = 0 \to 1$
Hybrid	Mean moderation	0.0782 to 0.0938	+19.9%	-0.235 to -0.266

Design	Architecture	Empirical SD, $t_{1/2} = 0 \to 1$	Change	Mean $\hat{\beta}_{bm:D}$, $t_{1/2} = 0 \to 1$
Hybrid	Covariance moderation	0.0708 to 0.0840	+18.7%	-0.231 to -0.239
OL+BDC	Mean moderation	0.0807 to 0.0956	+18.5%	-0.231 to -0.281
OL+BDC	Covariance moderation	0.0789 to 0.0904	+14.6%	-0.232 to -0.234

The changes are computed from unrounded SDs (verified by rerunning Appendix 1 on 2026-10-07; the mean-moderation Hybrid change, earlier given as 20.0%, is 19.9%).

Empirical SD grows as much or more under mean moderation. What distinguishes the architectures is the numerator: under mean moderation $|\hat{\beta}_{bm:D}|$ grows with carryover by 13% (Hybrid) and 22% (OL+BDC), with MCSE near 0.003, so the change is far outside Monte Carlo error. Under covariance moderation $\hat{\beta}_{bm:D}$ is stable.

Three consequences follow.

- The statement in Section 3.2 that the interaction coefficient "stays essentially unbiased under both architectures" (line 850) is false for mean moderation.
- Mean moderation's apparent robustness to carryover is, at least in part, an estimand shift. As carryover grows, the fitted `bm:Dbc` coefficient targets a larger quantity, which offsets the SE growth and holds power up. This is the mechanical consequence of the DGP/analysis mismatch that Section 2.2.1 itself identifies (lines 373-384): the mean-moderation DGP places no interaction signal off drug, while `Dbc` is nonzero off drug whenever carryover is present. (The direction of the shift is verified; the explanation is inferred.)
- The architecture contrast in Section 3.1 therefore compares one architecture whose target is fixed with another whose target moves with carryover. Power losses under the two are not losses of the same kind.

2.4 Test calibration accounts for a large share of the architecture gap

Evidence: verified (replicate reanalysis); size-adjusted figures are approximate.

The ratio κ of mean model-based SE (`betaSE`) to empirical SD of $\hat{\beta}_{bm:D}$ is far from uniform across cells.

Table 3: Ratio κ of model-based SE to empirical SD by design and architecture

Design	Mean moderation, $t_{1/2} = 0 \to 1$	Covariance moderation, $t_{1/2} = 0 \to 1$
CO	0.95 to 0.95	0.97 to 1.00
Hybrid	1.02 to 1.04	1.12 to 1.16
OL+BDC	1.10 to 1.20	1.12 to 1.26

Values are for the $c_{bm} = 0.45$ cells. The null cells show the same pattern (CO 0.92 to 0.98; OL+BDC 1.12 to 1.23).

The Type I error results are consistent with this.

- All six CO null cells lie above nominal (0.061 to 0.074; mean 0.067). CO is anti-conservative.
- Hybrid and OL+BDC null cells lie between 0.010 and 0.053, and the OL+BDC covariance-moderation null rate falls from 0.032 to 0.010 as carryover increases.

Power was recomputed using, for each cell, the 95th percentile of $|\hat{\beta}_{bm:D}/\widehat{SE}|$ in the matching null cell as the critical value. With 1,000 null replicates this quantile is itself noisy, so the figures below indicate magnitude only.

Table 4: Nominal and size-adjusted power loss by design and architecture

Design	Architecture	Nominal loss	Size-adjusted loss
CO	Mean moderation	0.014	0.043
CO	Covariance moderation	0.022	0.054
Hybrid	Mean moderation	0.058	0.021
Hybrid	Covariance moderation	0.129	0.050
OL+BDC	Mean moderation	0.021	0.028
OL+BDC	Covariance moderation	0.238	0.125

The covariance-minus-mean gap falls from 0.071 to about 0.029 in Hybrid and from 0.217 to about 0.097 in OL+BDC, and is about 0.011 in CO. Roughly half of the reported architecture gap in the two non-CO designs is attributable to the model-based test becoming progressively more conservative, not to a loss of information in the data.

The following manuscript statements are contradicted.

- Type I error is “uniformly sub-nominal” (line 927). The maximum is 0.074, and every CO cell exceeds 0.05.

- The miscalibration is "common-mode with respect to the architecture contrast" (lines 931-933). It differs by design, and within Hybrid it differs by architecture (1.12 against 1.02 at $t_{1/2} = 0$).
- "The within-replicate model SE matches it at scale" (line 944). The ratio spans 0.92 to 1.26.
- The companion calibration study's "6 to 10 percent" overstatement (line 931) understates the OL+BDC overstatement observed here.

The architecture difference in SE calibration may itself be a real and reportable finding. A homoscedastic residual model fitted to covariance-moderation data, whose conditional variance differs between on-drug and off-drug occasions, would be expected to misstate its standard errors. That is an architecture-specific effect, but it belongs to the analysis model, and it should be reported as such rather than as irreducible information loss. (Inferred.)

2.5 The design-based explanation contradicts the design specification

Evidence: inspected (driver lines 65-103); the alternative explanation is inferred.

The designs used for Section 3 are as follows.

Table 5: Occasions, spacing and off-drug occasions per path for each design

Design	Occasions (weeks)	Spacing	Off-drug occasions per path
CO	2.5, 5, …, 20 (8 occasions)	2.5 weeks	Path A: last 4 (10 weeks, after drug). Path B: first 4 (placebo-first, no carryover)
Hybrid	4, 8, 9, 10, 11, 12, 16, 20	1 to 4 weeks	Paths A/B: weeks 11-12 and one of 16/20. Paths C/D: weeks 10-12 and one of 16/20
OL+BDC	4, 8, 12, 16, 17, 18, 19, 20	1 week in BDC	Path A: weeks 19-20. Path B: weeks 18-20

The manuscript's account of design dependence (Abstract; Sections 3.1, 3.2 and 4.1; Conclusion 2) holds that OL+BDC "sustains its off-drug window the longest

of the three designs", that CO "alternates quickly enough" for little variance inflation to act, and that the OL+BDC blinded phase is "extended". None of these holds.

- CO path A has the longest uninterrupted post-drug off period of any path in any design: four occasions over ten weeks.
- CO does not alternate quickly. Each participant switches once.
- The OL+BDC blinded phase is four weeks long, with two or three off-drug occasions.

A more parsimonious explanation is measurement spacing relative to the carryover half-life. In CO the first off-drug measurement falls 2.5 weeks after discontinuation, when the residual fraction is $2^{-2.5} \approx 0.18$ at $t_{1/2} = 1$ and $2^{-5} \approx 0.03$ at $t_{1/2} = 0.5$. In Hybrid and OL+BDC the first off-drug measurement falls one week after discontinuation, when the residual fraction is 0.5 at $t_{1/2} = 1$. CO's off-drug occasions are therefore nearly uncontaminated at every half-life in the grid, which by itself accounts for its insensitivity. (Inferred; the residual fractions are verified by arithmetic.)

The competing explanation offered in the manuscript, that "within-subject AR(1) correlation already carries most of the signal" in CO, is not tested anywhere in the paper, and Section 3.2 concedes that it is "offered … in place of that missing derivation" (lines 901-902). The Abstract nonetheless presents it as established.

Recommendation. Replace the uninterrupted-off-drug-time account with an account based on the density of off-drug measurements within one or two half-lives of discontinuation, and test it directly, for example by varying CO spacing at fixed total duration.

2.6 The working-independence check is misreported

Evidence: verified (comparison of the manuscript's own figures).

Section 3.2 reports that summing $(D_{bc,it} - \bar{D}_{bc})^2$ predicts information losses of 36.8% (Hybrid), 4.6% (CO), and 51.2% (OL+BDC), and states that this "correctly orders the three … under either architecture" (lines 877-882).

The observed mean-moderation power losses are 6.9% (Hybrid), 1.9% (CO), and 2.8% (OL+BDC). The approximation places OL+BDC above Hybrid, while the data place Hybrid well above OL+BDC. The ordering therefore matches the covariance-moderation losses only. A predicted 51% information loss accompanied by a 2.8% power loss also contradicts the statement that design-contrast erosion "alone explains mean moderation's modest, design-dependent loss" (lines 836-837). Section 2.3 above supplies a likely reason: the mean-moderation estimand grows with carryover and offsets the lost contrast.

3. Major findings: internal consistency and estimand

3.1 The mean-moderation DGP is described two ways

Evidence: inspected.

Section 2.2.3 (lines 443-451) and Section 4.1 (lines 1007-1009) describe a mean-moderation DGP in which the interaction scales with exposure-decayed D, so that "the ratio … is preserved at every level of exposure" and "carryover keeps the same proportion off drug". The implementation applies the shift at on-drug occasions only, which Section 2.2.1 (lines 373-384) and Appendix B.6 state correctly. Section 2.2.3's first paragraph and the Section 4.1 passage should be rewritten to match the implementation, and the carryover-invariance claim for $\beta_1\beta_{bm}$ (line 485) should be withdrawn in light of Section 2.3.

3.2 True value, units, and sign convention

Evidence: inspected.

- Section 2.3 defines the reference true value as $\theta_{\text{true}} = -c_{bm}\sigma_{BR}$ (line 691). Section 2.2.3 gives the population slope as $c_{bm}\sigma_{BR}/\sigma_{bm}$ (written there with σ_B). The two differ by a factor of σ_{bm} and cannot both be the target of the unstandardized `bm:Dbc` coefficient.
- Bias relative to θ_{true} is listed as a performance measure but never reported.
- The sign convention required by NOTATION rule 5 (components are symptom reductions, so interaction coefficients are negative) is never stated. All DGP formulas in Section 2.2 are written with positive signs, while every reported estimate is negative.

3.3 Comparability of absolute power across architectures

Evidence: inspected.

Appendix B.8 shows that the two architectures have different marginal variances at a common c_{bm}, and concludes that "absolute power is not comparable between the two architectures" and that Section 3 compares only within-architecture profiles (lines 2280-2284). Section 2.2.1 states the opposite, that the common parameter value "licenses running both" (line 368), and the Abstract and Conclusions compare losses in absolute percentage points computed from different baselines. The manuscript should adopt one position. If cross-architecture comparison is retained, calibrating the architectures on realized biomarker-response correlation, or on power at $t_{1/2} = 0$, would make the comparison defensible.

3.4 Section 2.2.4 and Appendix B disagree

Evidence: inspected.

- "Four differences" names different lists in the two places. Section 2.2.4 lists correlation structure, gate, carryover (described as unchanged), and analysis model. Appendix B.7 lists A_c, $C_{cc'}$, b_p, and carryover (described as changed). Appendix B.7 then states that the cross-factor form "is not listed among the four differences of Section 2.2.4" while listing it in its own table of four.
- Section 2.2.4 (lines 617-622) and Appendix B.2 state that compound symmetry restricts c_{bm} more tightly than AR(1). Appendix B.7 shows that for any $t_{1/2} > 0$ the ordering reverses, with the published implementation admitting the higher ceiling. *Note (2026-10-07).* The reversal is a property of the published step coupling, which makes the coupling constant, and the interaction zero, at any positive half-life (`docs/37`). Under graded coupling ($g_t = 2^{-t_{sd}/t_{1/2}}$ off drug after exposure, 0 before it) the ordering does not reverse: at $\rho = 0.8$ the CS ceiling is 0.256 without carryover and at most 0.469 at $t_{1/2} = 2$, while the separable AR(1) ceiling is 0.440 (Hybrid, no carryover) and higher at every half-life and design (verified, `analysis/data/quick-sim/carryover-closed-form/ceilings-graded-by-rho.csv`; `docs/44`, Step 6).

3.5 Unfinished material and reproducibility gaps

Evidence: inspected.

- Three placeholders remain in the manuscript: `[TO RE-VERIFY]` at line 623, `[TO RE-RUN]` at line 630, and `[TO RE-VERIFY]` at line 2178.
- The Reproducibility section points to the Hendrickson comparison arm, whose results have been withdrawn, and does not identify the primary driver (`analysis/scripts/quick-sim/01-dgp-prototype.R`), its seed (`20260509`, with per-cell seeds `100000 * i + idx`), or the result files.
- Line 285 states that the three designs "are defined in Section 3". They are not defined anywhere in the manuscript beyond their names and path counts; the Hybrid schedule appears only in Appendix B.2. A design table equivalent to the one in Section 2.5 of this review is needed, and the design explanation cannot be assessed by a reader without it.

3.6 Conflicting statements about recoverability

Evidence: inspected.

Section 4.3 describes the fading correlation as "lost information no model can recover" (lines 1280-1281). Section 4.1 argues that a class-aware analysis could

in principle recover part of the loss (lines 1155-1159), and Section 2.2.3 argues that the population slope is unaffected. Section 4.3 also calls the shrinking exposure variance "recoverable", although no analysis restores contrast that the design does not provide. The passage should be reconciled with Sections 2.2.3 and 4.1, and with Section 2.4 of this review, which suggests that part of the loss is recoverable by better SE calibration.

4. Minor findings

Evidence: verified unless noted.

- **Abstract, line 80.** "$z = 2.80$, $p = 0.005$" is attached to the within-architecture covariance-moderation loss. It is the difference-of-differences statistic. The within-architecture z for that loss is about 7.0, while the adjacent mean-moderation "$z = 3.34$" is within-architecture. The two parallel sentences report different statistics.
- **Line 703.** "The highest cell in the grid is 0.842" should read 0.844 (mean moderation, Hybrid, $t_{1/2} = 0.5$).
- **Lines 780-782.** The step from 0.754 to 0.723 is not "within one MCSE". The SE of the difference is about 0.020, so the step is about 1.6 SE.
- **Appendix B.8, line 2248.** The conditional SD $8\sqrt{1 - 0.45^2} = 7.144$ should read 7.14, not 7.15. The statement that a simulation "reproduces each to three decimal places" should be checked against the corrected value.
- **Line 410.** The limit $\lambda = 0$ corresponds to an infinite half-life, not to a "no-carryover model variant".
- **Line 749 and elsewhere.** OL+BDC is called a "two-period design". It is a two-path design.
- **Lines 1324-1327.** The closing sentence of Section 4.4 is logically inverted. If the architecture choice matters least in parallel-group designs, assuming it harmless there is not wrong. The intended target is presumably within-subject designs.
- **Line 982.** "Essentially immune to carryover" conflicts with a statistically significant 6.9% loss in the focal Hybrid design ($z = 3.34$).
- **Line 189.** "As Section 4 shows" should refer to Section 3.2.
- **Uncited or unsupported claims.** The following need a citation, a supporting analysis, or removal.
  - $N = 70$ as "the smallest of the prazosin-PTSD-matched sample sizes used in the published methodological literature" (lines 700-701).
  - "Methodological work on N-of-1 carryover sharing authors with Hendrickson et al." using mean moderation (lines 1301-1304).
  - "A companion paper on carryover-mitigation strategies" (lines 1289-1290).

  - The "8-13%" and "40-60%" ranges from "earlier exploratory runs" (lines 751-753, 783-785, 1334-1336). These are unpublished and their removal is recommended.
  - Parallel-group enrichment designs giving "similar power under either architecture" (lines 1315-1316). This is untested and in tension with Appendix B.8.

- **Rendered PDF numbering.** The table of contents reads "1 Abstract", "2 1. Introduction", "3.1 2.1 Notation and analysis model". The cause is `number_sections: true` combined with hand-numbered `\section{}` and `\subsection{}` titles. Either use `\section*{}` with manual numbers or remove the manual numbers.
- **Line 1272.** `\end{itemize} These requirements ...` begins a new paragraph on the closing line of the list; a blank line is needed.
- **Appendix B.5.2 (inspected).** The loop-order dependence of the cross-factor decay rate is a code defect at `R/generateData.R:334`, where `rho` is taken from the outer-loop factor and the later of the two writes survives. It should be fixed in the package rather than documented as an artifact.
- **Appendix B.7.** The smallest-eigenvalue expression $(1 - \rho_c) - (c_1 - c_\times)$ assumes a common ρ_c across factors, which should be stated.
- **Appendix B.4 and B.5.3.** "At any positive half-life b^{orig} is constant" holds only when every off-drug occasion follows a drug exposure. It fails for placebo-first paths such as CO path B.

5. Notation

Assessed against `analysis/report/NOTATION.md`. The linter passes, so the items below concern rules it does not mechanize.

The symbols discussed in this section are quoted from the manuscript (see the notation index).

- b **is overloaded.** It denotes a realized biomarker value (Sections 2.2.2 and 2.2.3), the standardized biomarker b_i (Section 4.1, consistent with NOTATION), and the biomarker coupling vector (Appendix B). Sections 2.2.1 and B.6 use z_i for the standardized biomarker where NOTATION specifies b_i. The coupling vector should be renamed, for example to g_t as in `docs/44`, or to r_{BM}. (This review earlier suggested γ; that symbol is reserved by NOTATION for the $BR \times PB$ interaction.)
- **Biomarker moments.** μ_B and σ_B (main text) and μ_{bm} and σ_{bm} (Appendix B.6 notes, B.8) are used for the same quantities, as are B_i and BM.
- **Estimand label.** $\beta_{bm:D}$ (Section 2.3) and $\hat{\beta}_{bm:D_{bc}}$ (Section 3.3) denote the same coefficient. NOTATION specifies $\beta_{bm:D}$.

- **Shared c_{bm} label.** The shared label is declared (line 360), which satisfies rule 2. However, Section 2.2.1's display uses β_{bm} and Section 2.2.3 mixes $\beta_1\beta_{bm}$ with c_{bm}, so the declaration does not hold throughout.
- **Architecture labels.** "Architecture B" appears first at line 547 and throughout Appendix B, but the main text never introduces the A/B/C labels given in the NOTATION table.
- **Time index.** t is both the occasion subscript and the time covariate in `Sx ~ bm + t + Dbc + bm:Dbc`, and Appendix B switches to p for occasions.

6. Companion files

6.1 `appendices.Rmd`

Evidence: verified (diff). The file is a verbatim copy of `report.Rmd` lines 1449-2292; the diff differs by a single blank line. It is untracked. Maintaining two copies guarantees drift, which is already visible elsewhere in this directory. Either include a single appendix source in both documents as a knitr child document, or drop the standalone file and render the appendices from `report.Rmd` alone. Every finding in Sections 3.4, 4 and 5 that concerns Appendix A or B applies equally to this file.

6.2 `bullets.Rmd`

Evidence: inspected. The report describes this file as its supplementary structured summary (lines 1438-1440), but it is out of date in several respects.

- Its title and its description of the companion manuscript's title do not match the report's title, "Data Generation Machinery for N-of-1 Clinical Trial Simulations".
- Its section numbering does not follow the report. It places the architectures at 2.1 and 2.2 rather than 2.2.1 and 2.2.2, refers to `Dbc` as defined in "Section 3.1", and places the dual-channel architecture in "Section 2.2".
- Its Section 3.3 (lines 218-240) reports the $N = 140$ OL+BDC robustness check with specific figures. The report states that this check "is not reported here" (line 1430).
- Its Section 3.2 describes the superseded account in which the signal is "attacked from both sides" through correlation decay, not the report's current precision account (which is itself contested in Section 2.2 of this review).
- Its Section 3.1 heading describes the covariance-moderation loss as "absent at CO", against the reported 3.0%.
- It omits the `in_header: ../sim-preamble.tex` include used by the other two files.

The file should be regenerated from the report once the revisions in Section 7 are complete, not edited in parallel.

7. Recommended order of work

1. Re-run the Section 3.1 factorial on the current package, recording the commit hash, and confirm convergence and null-cell rates.
2. Add to the reported performance measures the ratio of model-based SE to empirical SD, size-adjusted power, and bias against a true value whose units match the fitted coefficient (Sections 2.4 and 3.2).
3. Rewrite the mechanism in Sections 2.2.3, 3.2 and 3.3 around what the data support: the estimand shift under mean moderation (Section 2.3), architecture- and design-dependent SE calibration (Section 2.4), and off-drug measurement density relative to $t_{1/2}$ (Section 2.5). Test the spacing explanation directly.
4. Add a design-specification table and complete the Reproducibility section (Section 3.5).
5. Resolve the internal inconsistencies of Sections 3.1, 3.3, 3.4 and 3.6, the minor items of Section 4, and the notation items of Section 5.
6. Revise the Abstract, Discussion and Conclusions to match.
7. Consolidate `appendices.Rmd` into a single source and regenerate `bullets.Rmd`.

8. What this review did not do

- No new simulation was run. The effect of the carryover correction on Section 3 (Section 2.1) is inferred in direction and unknown in magnitude.
- The size-adjusted power figures use null-cell quantiles from 1,000 replicates and were not bootstrapped; they indicate magnitude only.
- The spacing-based explanation of design dependence (Section 2.5) is a hypothesis consistent with the design specification and the residual fractions; it has not been tested.
- The Section 3.2 working-independence calculation was not rerun; only its reported figures were compared with the power table.
- The positive-definiteness ceiling of Appendix B.7, the two-million-draw check of Appendix B.8, and the Hendrickson comparison arm were not re-executed.
- Citations were not checked against `references.bib` or against the cited sources.
- Prose style was not reviewed beyond the specific wording problems listed in Section 4.

Appendix 1. Reanalysis script

```
library(data.table)
x <- readRDS(file.path(
  'analysis/data',
  'quick-sim/01-dgp-replicates.rds'
))
x[, z := beta / betaSE]
s <- x[, .(
  mean_beta = mean(beta),
  emp_sd = sd(beta),
  mean_model_se = mean(betaSE),
  se_ratio = mean(betaSE) / sd(beta),
  power = mean(p < 0.05)
), by = .(architecture, design, t1half, c.bm)]
setorder(s, architecture, design, c.bm, t1half)
print(s, digits = 3)

crit <- x[c.bm == 0, .(crit = quantile(abs(z), 0.95)),
  by = .(architecture, design, t1half)]
alt <- merge(x[c.bm == 0.45], crit,
  by = c('architecture', 'design', 't1half'))
adj <- alt[, .(
  nominal = mean(p < 0.05),
  size_adj = mean(abs(z) > crit[1])
), by = .(architecture, design, t1half)]
setorder(adj, architecture, design, t1half)
print(adj, digits = 3)
```

Run from the repository root with `Rscript -e "source('path/to/script.R')"`.